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Three new species in the endemic New Zealand genus *Alloecentrella* (Trichoptera), and a re-evaluation of its family placement

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Abstract

Three new species of caddis in the endemic New Zealand genus *Alloecentrella* Wise, 1958 (Trichoptera: Sericostomatoidea) are named and described: *A. linearis* sp. n., *A. cirratus* sp. n., and *A. incisus* sp. n. We report the known distribution of each of the four species in the genus; *A. magnicornis* and *A. incisus* appear to be found only in North Island; *A. linearis* and *A. cirratus* have been recorded only from South Island. We provide characters to differentiate the larvae (extent of the head carina; the shape of the head capsule in lateral profile) and cases (position of the terminal aperture). We present the results of a phylogenetic analysis of larval, pupal and adult morphological characters which places *Alloecentrella* in the Helicophidae.

Keywords: Calocidae, Helicophidae, *Alloecentrella*, new species, New Zealand, Trichoptera, taxonomy, distribution

Introduction

In the first report of his work on the New Zealand Trichoptera, K. A. J. Wise (1958) presented a catalogue of the Auckland Museum Trichoptera collection, largely formed from the CE Clarke 1929 bequest. His study disclosed new genera and species, such as *Alloecentrella* and its type species *A. magnicornis*, based on a male adult from [AK] Waitakere, 28 October 1934, collector unknown. Wise placed *Alloecentrella magnicornis* in the Beraeidae despite requiring an extension to that family's definition regarding wing venation (presence of a discoidal (=radial) cell in the forewing). He also stated that it could be placed in the Helicophidae, but not in either of its existing genera, and noted the general similarities in wing venation and male genitalia in these two families. His choice of Beraeidae was based on other (unspecified) characters but he noted that the male genitalia of *A. magnicornis* was of similar form to that of the Australian species *Alloecella* “*warneri*” [correctly: *warneria*] Mosely, 1953, at that time placed in the Beraeidae along with the New Zealand genus, *Pycnocentrella* Mosely, 1953.

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The venational patterns of *Alloecella* and *Alloecentrella* males are unusual, due to the loss or fusion of veins, and homologies are not immediately obvious. McFarlane (1966) described the females of *Alloecentrella magnicornis* and *Pycnocentrella eruensis* Mosely, 1953 (which have more complete venation compared to the males), and expressed “grave doubts as to the propriety of including them in the Beraeidae”. When Ross (1967) erected a new family Pycnocentrellidae he did not indicate what taxa it contained but it can be assumed he intended it to contain the monotypic genus *Pycnocentrella*. Wise (1973) listed both *P. eruensis* and *A. magnicornis* in the Pycnocentrellidae. Neboiss (1977) moved *Pycnocentrella* to the Australian family Calocidae, but made no mention of *Alloecentrella*. When describing the larva of *Alloecentrella magnicornis*, Cowley (1978) formally transferred the genus to the Helicophidae, based on wing venation, the arrangement of thoracic setal warts and larval similarities to the genus *Zelolessica* McFarlane, 1956.

Wiggins (1982) listed *Alloecentrella* in the Helicophidae but Neboiss (1986) again placed it in the Calocidae and Winterbourn et al. (2006) have adopted Neboiss’ placement as does Morse (2006). Neboiss (2002) described a new genus, *Heloccabus*, placing it provisionally in the Helicophidae and providing a matrix of 16 larval, pupal and adult characters for some members of Helicophidae, Calocidae and related families. We re-analyse this character matrix, extended to include New Zealand taxa, to show that *Alloecentrella* belongs in the Helicophidae.

This paper formally names and describes adults of the three *Alloecentrella* species noted by Henderson (1985), gives characters to distinguish their cases and larvae, and reports on the known distribution of the genus. Of the four species, *A. magnicornis* appears to have the most extensive distribution, covering almost all of North Island from Northland to Wellington, Taranaki to Bay of Plenty (data from the New Zealand Trichoptera Database, Ward & Henderson (1993)). There are three published South Island location records of *A. magnicornis*, from BR and WD (Cowley 1978; Collier et al. 1989). However, these records are of larvae only and, based on geographic location, are most likely to belong to *A. linearis*. Both *A. linearis* and *A. cirratus* appear to be restricted to South Island, and *A. incisus* to North Island, New Zealand (Figure 9).

The adults of all four species of *Alloecentrella* are small (forewing length 3.5–7 mm), black and day-active. Most specimens are caught by sweep-netting streamside vegetation and very few have been captured in light traps. The larvae are usually associated with bryophytes, feeding on the plants and using leaf material in their cases.

Materials and methods

Specimen repositories and other abbreviations

ALEX: Alexandra Museum, Alexandra, New Zealand; CMNZ: Canterbury Museum, Christchurch, New Zealand; AMNZ: Auckland Institute and Museum, Auckland, New Zealand; IHPC: Ian M Henderson’s personal collection, Massey University, Palmerston North, New Zealand; NIWA: NIWA Ecosystems, PO Box 11-115, Hamilton, New Zealand; NZAC: New Zealand Arthropod Collection, Landcare Research, Auckland, New Zealand; OMNZ: Otago Museum, Dunedin, New Zealand; SMNH: Swedish Museum of Natural History, Stockholm, Sweden.

Storage medium. alc = 80% SDA (industrial ethanol); pin = dry pinned.

Collector’s names. George W Gibbs = GWG; Ian M Henderson = IMH; Brian H Patrick = BHP; John B and Geraldine M Ward = J&G; John B Ward = JBW.

Two-letter codes for regions in New Zealand (Crosby et al. 1998). AK: Auckland; BR: Buller; CL: Coromandel; GB: Gisborne; HB: Hawkes Bay; MB: Marlborough; MC: Mid-Canterbury; NC: North Canterbury; ND: Northland; NN: Nelson; RI: Rangitikei; SD: Marlborough Sounds; TK: Taranaki; TO: Taupo; WA: Wairarapa; WD: Westland; WN: Wellington.

Grid references to collection localities are in terms of the NZMS 260 metric grid, to a precision of 100 m. Altitudes are given in metres (m) above sea level. Nomenclature for wing venation follows Mosely (1953).

Systematic account

Genus *Alloecentrella* Wise, 1958

Diagnosis (based in part on Wise (1958), McFarlane (1966) and Cowley (1978)).

Adult. Ocelli absent, tibial spurs 2,2,4, maxillary palps 5-segmented in both sexes, pronotum with a single elongate wart, mesoscutum without warts, scutellum with a single round wart. Wing coupling by a row of curved macrotrichia on the costal margin of the hind wing and folded anal margin of the forewing.

Wing venation reduced with vein homologies uncertain. Forewing radial cell closed. Cubitus apparently absent in male forewing, present in the female but confluent with a branch of the media. In the hind wing the cubitus (?) forks twice forming three terminal veins. Radial cell absent in male, closed in female. In the male the media of the hind wing consists of two veins diverging gradually or partially fused.

Male. Tergite X bifid and bearing two pairs of thick setae dorsally. Phallus partially enclosed by a compound penis cover which is “U” shaped in cross-section but with dorsal inner flanges almost forming a tube. Inferior appendages single jointed, inner aspect concave, with dorsal and ventral branches turned inwards apically. Segment IX narrow dorsally and ventrally.

Female. Abdomen terminating in a genital chamber formed by dorsal and lateral extensions of tergite IX, fringed with setae curling inwards. Ventrally a pair of projecting rounded processes flanked by raised striated ridges terminating in acute projections.

Larva. Case curved, tubular, mainly of bryophyte fragments. Antennae approximately halfway between eye and base of mandible. Ventral apotome an elongate triangle separating the genae. Pronotum fully sclerotised, mesonotum sclerotised on the anterior 2/3 of the segment, metanotum with a pair of thin plates bearing setae. Abdominal segment I with a small raised dorsal tubercle and lateral pads of spicules. No gills or abdominal hair fringe but a row of spicules on segment VIII. Anal claw with a single small accessory claw.

Descriptions of the adults

Alloecentrella magnicornis Wise, 1958 (Figure 1)

Type material. Holotype ♂ (AMNZ, genitalia in alcohol), AK, Waitakere, 28.X.1934 (anon.).

Other material examined (all IHPC, alc). 3 ♂, ND, Mangamuka R at Soda Springs, 25550, 66672, 60 m, IMH, 9.I.2006. 2 ♂, 1 ♀, ND, Mangamuka R trib. at Soda Springs, 25548, 66671, 80 m, IMH, 9.I.2006. 3 ♂, ND, Waipapa R, 25812, 66666, 110 m, IMH, 29.XII.1995. 2 ♂, ND, Puketona Stm,

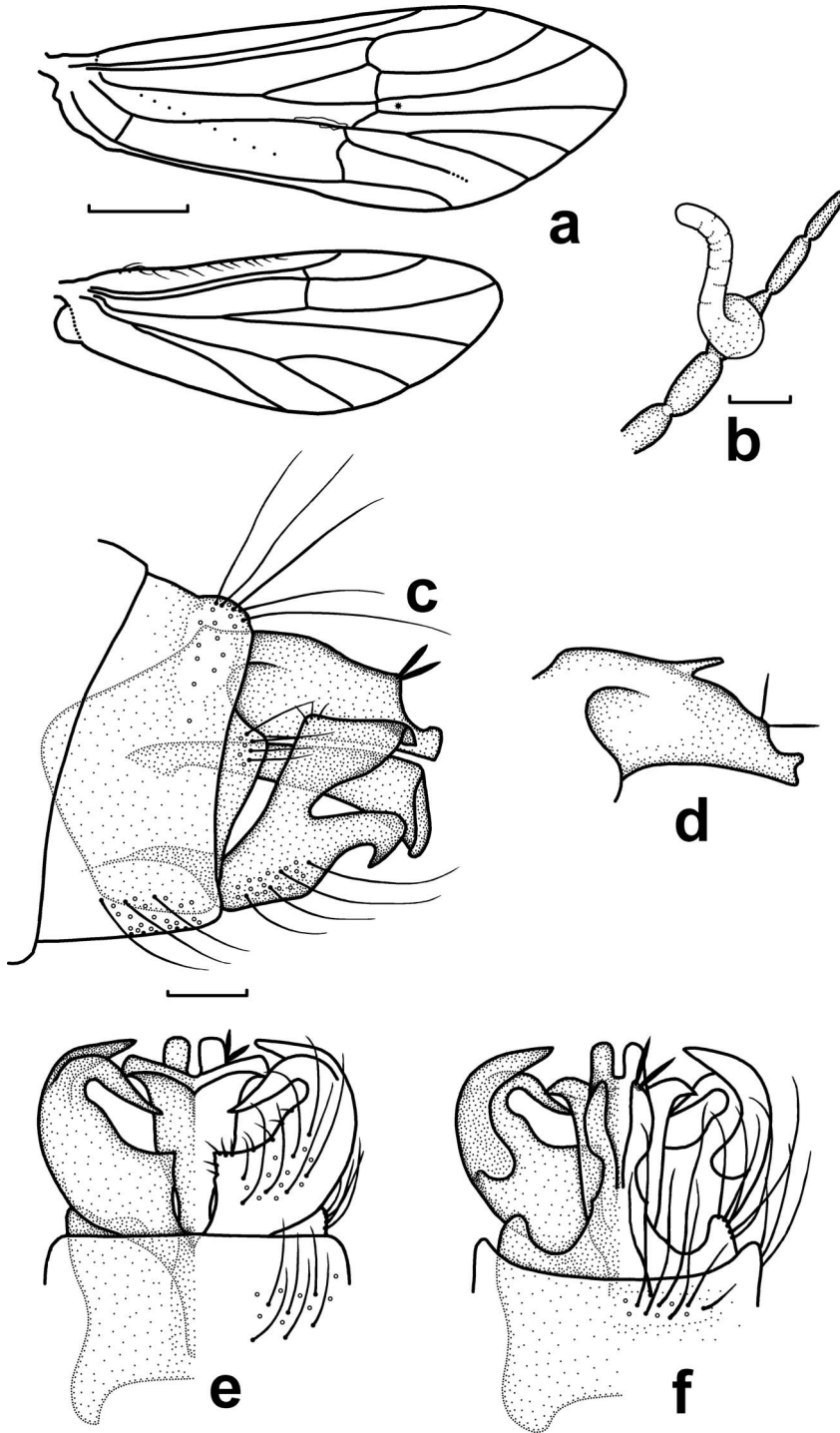


Figure 1. *Alloeocentrella magnicornis* male (a) wings (scale bar=1 mm); (b) maxillary palp; (c) genitalia lateral; (d) variation in segment X; (e, f) genitalia ventral, dorsal (scale bar=0.1 mm).

Pokaka, 25970, 66581, 120 m, IMH, 24.XII.1999. 3 ♂, ND, Punaruku Stm, 26301, 66443, 20 m, IMH, 2.I.1996. 1 ♂, ND, Waikohatu Stm, Waipoua Forest, 25615, 66196, 300 m, IMH, 20.I.1986. 4 ♂, AK, Waitaraire Stm, 26523, 65378, 60 m, IMH, 14.I.2006. 2 ♂, AK, Waitangi Falls, Omeru, 26425, 65149, 10 m, IMH, 30.XII.1993. 3 ♂, 1 ♀, AK, Hellyer Rd, Kaukapakapa, 26468, 65113, 60 m, IMH, 24.XII.2006. 5 ♂, 1 ♀, AK, Fairy Falls, Waitakere Ra., 26490, 64753, 200 m, IMH, 5.I.1987. 1 ♂, larvae, CL, Waiora Stm, 27452, 64576, 140 m, IMH, 13.II.1992. 3 ♂, 1 ♀ (pair in copula), CL, Marorauririki Stm, 27469, 64551, 120 m, IMH, 12.II.1992. 1 ♂, GB, Mangapuwerawera Stm, 28723, 62663, 760 m, IMH, 21.II.1981. 2 ♂, HB, Boundary Stm at Shine Falls, 28432, 62263, 360 m, IMH, 17.I.1998. 4 ♂, 1 ♀, HB, Boundary Stm, upper bridge, 28411, 62255, 580 m, IMH, 18.I.1998. 1 ♂, TK, Timaru Stm, Pukeiti, 25945, 62213, 410 m, IMH, 18.XI.2001. 1 ♂, TO, Omoho Stm., Mt. Tihia, 27462, 62428, 960 m, IMH, 17.II.1983. 1 ♂, TO, Mangatawai Stm, Desert Rd., 27490, 62239, 800 m, IMH, 8.I.1987. 2 ♂, TO, Boyd Hut, Ngaruroro R., 27839, 62229, 1040 m, IMH, 17.I.1982. 3 ♂, 1 ♀, RI, Mokai-Patea Range, 27738, 61641, 900 m, IMH, 14.I.1982. 1 ♂, WN, Kahuterawa Stm.trib, 27305, 60778, 260 m, IMH, 12.I.1993. 2 ♂, 1 ♀, WN, Otangane Stm, 27340, 60772, 440 m, IMH, 2.I.2004. Larvae, WN, Burn Hut, Tararua Ra, 27244, 60639, 670 m, IMH, 7.IV.1981. 3 ♂, 1 ♀ (pair in copula), WN, Waiotauru R, 26969, 60337, 150 m, IMH, 15.XII.1998. 1 ♂, WN, Waiotauru R, trib., 26960, 60318, 220 m, IMH, 13.XII.1997, 3 ♂, IMH, 21.I.1999. Larvae, WN, Atiwhakatu Stm., 27178, 60308, 340 m, IMH, 24.III.1981. 3 ♂, 1 ♀, WN, Catchpool Stm, 26730, 59832, 100 m, IMH, 2.I.1998. 2 ♂, WN, Gollans Stm, 26690, 59865, 30 m, GWG, 30.I.1980.

Description

Male (Figure 1). Maxillary palps 5-jointed, a thin-walled, setose, eversible scent organ on segment 3 (Figure 1b).

Wing venation (Figure 1a). Forewing, Sc almost straight, no cross-veins except the humeral. Radial cell about 1/5 of wing length, closed by a slightly concave cross-vein. Apical forks 2 and 3 present, base of fork 2 is a cross-vein below the radial cell, nygma at base of fork 2. Media branches twice to form m1, m2 and m3 + 4, the latter joined by a sinuous cross-vein to the single anal vein. A small hyaline area before the first branching of the media. Cubitus apparently missing entirely. An oblique fold in the forewing across the basal third of the media. Hind wing radial sector missing. Media forks once at about 1/3 of wing length, the two branches almost parallel over the middle third of the wing. Cubitus branches at a shallow angle near the base and at about half the wing length. A row of about 13 curved macrotrichia along the costal margin. Forewing length 4.7–6.4 mm (mean = 5.6, sd = 0.47, $n = 50$).

Genitalia. Segment X narrows at midpoint where two dorsal ridges diverge. A pair of strong black spines arise from a raised point on each dorsal ridge close to the apex which is excised dorsally and bifid below. (In some individuals the dorsal margin tapers more gradually towards the apex, there is a pair of narrow projections at the midpoint and the pairs of spines are longer and thinner (Figure 1d). Both forms can be collected from throughout the species range and at the same location and time.) The penis cover widens and is flared downwards posteriorly with a recurved ventral lip. Inferior appendage with lower branch about halfway along its length, strongly recurved with acute apex. Upper branch also strongly inturned and acute, triangular in lateral view. Many long black setae arise from the base of the inferior appendages and from segment VIII.

Female. Wing venation as for *A. linearis* (Figure 3a) except that the forewing radial cell is a little longer. McFarlane (1966) illustrates a specimen in which in the hind wing the media

does not fork and the radial cell is not closed. Forewing length 5.6–6.8 mm (mean = 6.5, sd = 0.41, $n = 7$).

Diagnosis

Males are easily distinguished from other members of the genus by the in-turned, spine-like apices of both branches of the inferior appendages. The maxillary scent organ is also found in *A. cirratus* n.sp.

Known distribution

Widespread in North Island only (Figure 9).

Alloecentrella linearis sp. n. (Figures 2 and 3)

Type material. Holotype ♂, paratypes 2 ♂, 1 ♀ (CMNZ, alc), 2 ♂, 1 ♀ (NZAC, alc), 2 ♂, 1 ♀ (ALEX, alc), 3 ♂, 1 ♀, 1 larva, 1 exuviae, (IHPC, alc), NN, Moa Park, Abel Tasman National Park, 25046 60298, 1000 m, IMH, 29.XI.1979. 1 ♂ paratype (IHPC, alc), BR Arnold River tributary, Lake Brunner, 23842 58471, 90 m, IMH, 13.XI.2002. 1 ♂ paratype (IHPC, alc), NN Cobb Reservoir, 24764 60083, 820 m, GWG, 20.XI.1979. 1 ♂ paratype (CMNZ, alc), BR Mole Stream tributary, swept, 24710 59139, 630 m, J&G, 4.XII.1993. 2 ♂ paratypes (IHPC, alc), BR Sabine River, Forks - Blue Lake, 24820 59080, 900 m, IMH 3.I.1999. 1 ♂ paratype (NZAC, pin), NN Mount Arthur, 4000 ft, A Philpott, 23.XII.1921. 2 ♂ paratypes (NZAC, pin), NN Nelson, A Philpott, 22.XII.1921. 6 ♂ paratypes (OMNZ, pin), 5 ♂ paratypes (CMNZ, alc), WD Okarito R.tributary, 22864 57655, 80 m, BHP and JBW, 6.XI.1995. 1 ♂ paratype (IHPC, alc), WD Lake Kaniere, Sunny Creek, 23569 58210, 140 m, IMH, 19.XI.1994. 1 ♂ paratype (CMNZ, alc) BR Lake Rotoiti, Kerr Bay, at light, 24973 59334, 620 m, J&G, 1.XII.2001. 1 ♂ paratype (IHPC, alc), NN Flora Track, Takaka Track Junction, 24832 60047, 780 m, IMH, 2.XII.1986. 1 ♂ (IHPC, alc), NN Growler Creek, Flora Track, 24826 60043, 860 m, IMH, 6.I.1981.

Other material examined. Larvae (IHPC, alc), NN Knuckle Hill, Whanganui Inlet, 24726 60626 320 m, IMH, 13.II.1980. Larvae (IHPC, alc), NN Heaphy north of Perry, 24633 60364, 460 m, IMH, 13.I.1981. Larvae (IHPC, alc), WD Deep Creek, Okarito, 22827 57692, 40 m, D Eward & R Putz, 11.I.1986. Larvae (IHPC, alc), WD Omoeroa Saddle, 22760 57515, 320 m, D Eward & R Putz, 11.I.1986. 1 ♂, larvae (IHPC, alc), NN Flora Stream, 24876 60028, 910 m, IMH, 6.I.1981. Larvae (IHPC, alc), NN Leslie River, 24740 59981, 300 m, IMH, 26.III.1983. Larvae (IHPC, alc), MB Tinline River, 25519 59926, 120 m, IMH, 18.II.1980.

Description

Male (Figure 2). Venation of forewing as for *A. magnicornis* (Figures 1a and 2a). Hind wing with medial veins confluent for 2/3 of wing length, forming a groove which encloses a single row of androconia. The coalesced veins and androconia form a distinctive line in the wing. Forewing length 4.9–6.5 mm (mean = 5.8, sd = 0.34, $n = 22$).

Genitalia. Segment X narrow basally, widens slightly to point of furcation at 3/4 of length. Dorsal surface with a pair of small protrusions, inclined downwards posteriorly to apex which is acute in lateral view. Penis cover lower margin curving downwards, rounded distally, dorsolateral projections rounded, triangular, close to distal end. Inferior appendages; proximal half directed outwards, distal half turned inwards, apex curved further inwards and acute, in lateral view tapering little and rounded distally. A small, acute ventral projection from the inferior appendage at about three quarters of its length distally.

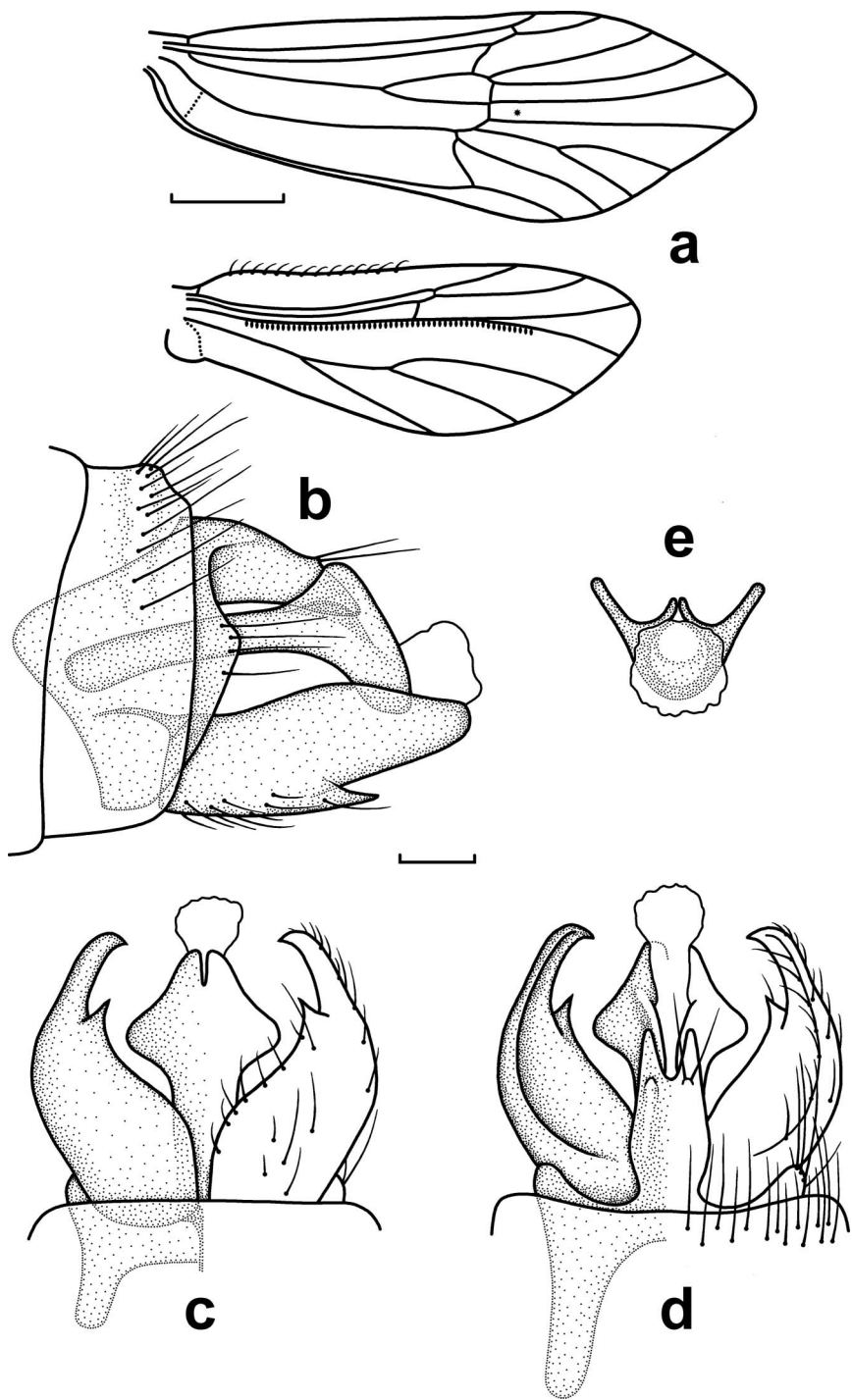


Figure 2. *Alloecentrella linearis* male (a) wings (scale bar = 1 mm); (b, c, d), genitalia lateral, ventral, dorsal; (e) penis cover, cross-section (scale bar = 0.1 mm).

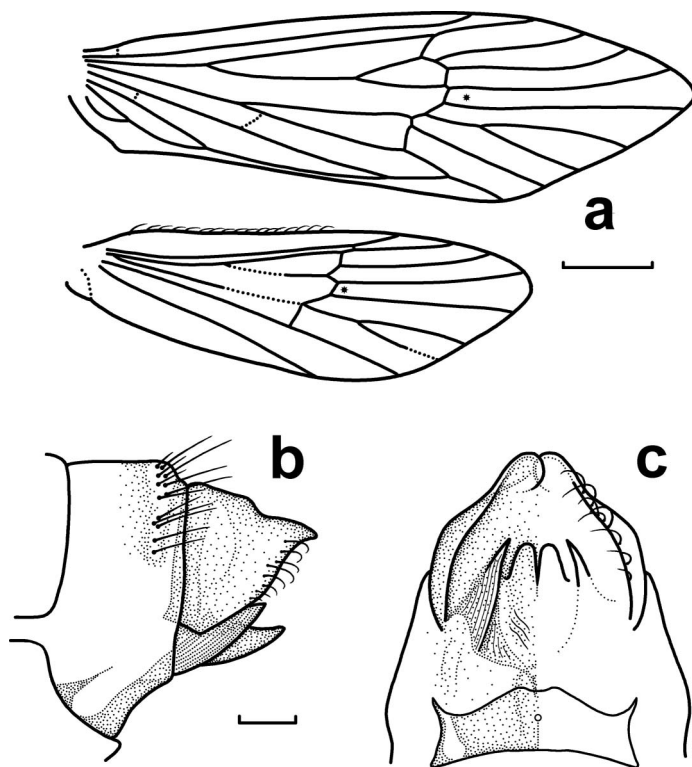


Figure 3. *Alloecentrella linearis* female (a) wings; (b, c) genitalia, lateral, ventral.

Female (Figure 3). Venation does not differ consistently from that of *A. magnicornis*. Forewing length 6.0–7.5 mm (mean = 6.9, sd = 0.66, $n = 4$).

Genitalia differ a little from *A. magnicornis* in that the medial ventral pair of lobes is thicker with more rounded apices.

Diagnosis

Males are easily distinguished by the genitalia, and by the distinctive line in the hind wing. *A. linearis* is the largest member of the genus.

Known distribution

Northwest of South Island only (Nelson, Buller and Westland) (Figure 9).

Etymology

Latin *linearis*, linear. The name alludes to the line in the male hind wing.

Alloecentrella cirratus n.sp. (Figure 4)

Type material. Holotype ♂, 1 ♂ paratype (CMNZ, alc), 2 ♂, 1 larva, (IHPC, alc), SD Waitohi Stream, Picton, 25938 59874, 30 m, IMH, 6.XII.1980. 1 ♂ paratype (NZAC, alc), SD Furneaux Lodge, north

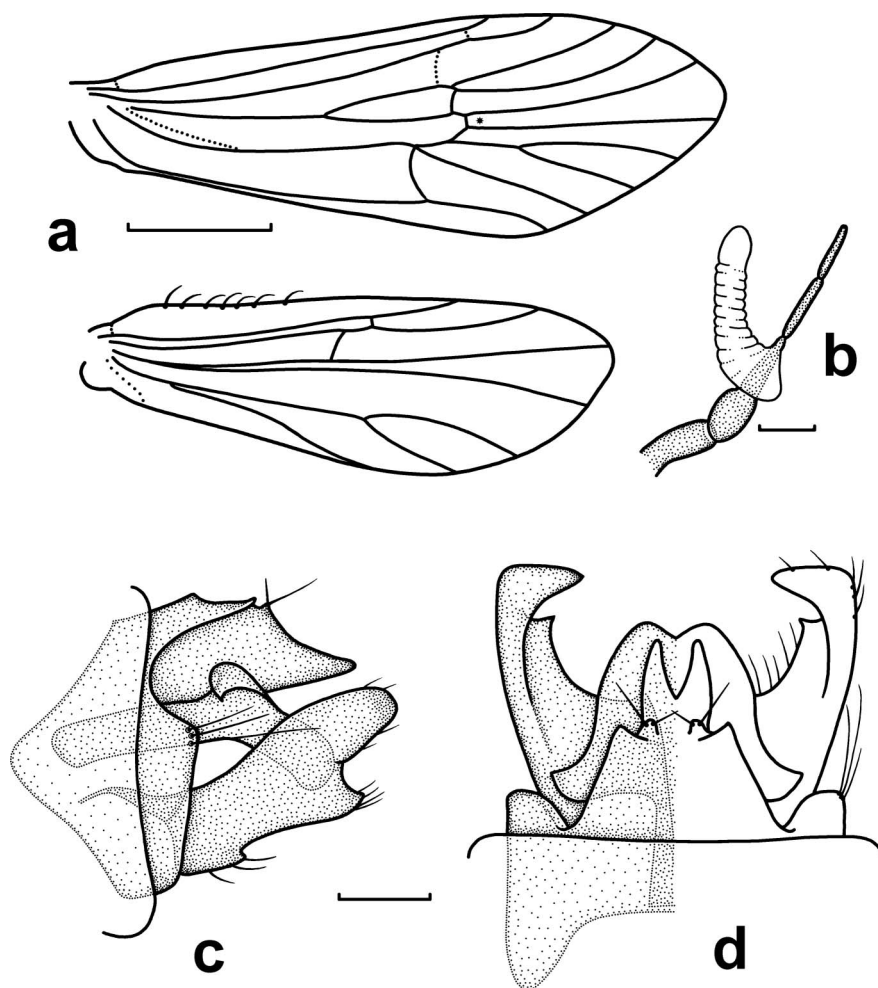


Figure 4. *Alloeocentrella cirratus* male (a) wings; (b) maxillary palp; (c, d) genitalia, lateral, dorsal.

stream, 26100 60121, 20 m, IMH, 17.I.2001. 1 ♂ paratype (IHPC, alc), SD Furneaux Lodge, waterfall, 26107 60122, 200 m, IMH, 16.I.2001. 2 ♂ paratypes (IHPC, alc), SD Resolution Bay, Ship Cove Track, 26128 60102, 100 m, IMH, 15.I.2001. 1 ♂ paratype (IHPC, alc), SD Ship Cove, main stream, 26133 60117, 40 m, IMH, 15.I.2001. 1 ♂ paratype (IHPC, alc) SD Ship Cove, waterfall, 26137 60124, 20 m, IMH, 15.I.2001. 2 ♂ paratypes (IHPC, alc), MB Pelorus Bridge, 25579 59897, 60 m, GWG, 13.XI.1979. 1 ♂ paratype (SMNH, alc), NC Greyneys Creek, 1 km N., 23952 58019, 680 m, KA Johanson & JBW, 2-7.I.2005, malaise trap.

Other material examined. Larvae (CMNZ, alc), MB Dog Stm., Waterfall Track, 24971 58560, 490 m, JBW, 30.V.1991. Larvae (CMNZ, alc), MB Dog Stm., trib., Mullans Rd., 24977 58559 440 m, JBW, 30.V.1991. Larvae (CMNZ, alc), MC Thomas R., trib., 24041 57772 800 m, JBW, 9.V.1991.

Description

Male. Maxillary palps with a thin-walled eversible structure present on segment 3. Forewing venation as for genus with the addition of a short vein fusing with the medial

vein near the wing base. Hind wing medial veins diverging from the wing base, running nearly parallel through central part of wing, no androconia. First branching of cubitus vein also very attenuated. Forewing length 4.2–5.4 mm (mean = 4.8, sd = 0.31, $n = 11$).

Genitalia. Tergite X broadly based, narrowing abruptly with small dorsal protrusions, distal 1/3 divided, apices acute and directed posteriorly. Dorso-lateral extensions arise midway along the penis cover, rolled outwards and downwards, acute. Inferior appendages' outer margin almost straight in ventral view, apex turned inwards at right angle and rounded; ventral margin with a small angled projection at one fifth of its length and a larger in-turned projection at three quarters of its length.

Female. unknown.

Diagnosis

Males have an eversible structure on the maxillary palps (scent organs). This structure is also found in males of *A. magnicornis* but not the other two species of *Alloecentrella*. The genitalia, particularly the curled lateral extensions of the penis cover and the sharp in-turned angle of the inferior appendages, are distinctive.

Known distribution

Northeast South Island only (Marlborough and Canterbury) (Figure 9).

Etymology

Latin *cirratus*, curled. The species is named for the curled form of the penis cover flanges.

Alloecentrella incisus n.sp. (Figures 5 and 6)

Type material. Holotype ♂, 2 ♂, 1 ♀ paratypes (CMNZ, alc), 6 ♂ paratypes (NZAC, IHPC, alc), RI Limestone Creek near Sixtus Lodge, swept, 27668 61324, 560 m, IMH, 22.I.2000. 3 ♂ paratypes (IHPC, alc), TK Waiweranui Stream, Egmont National Park 25931 62151, 430 m, IMH, 16.XII.1980. 2 ♂, 1 ♀ paratypes (IHPC, alc), HB Boundary Stream at Shine Falls, 28432 62263, 360 m, IMH, 17.I.1998. 2 ♂, 3 pupal paratypes (IHPC, alc), RI Waitotara River, stream near Tawhiwhi, 26684 61716, 40 m, IMH, 22.XII.2000. 9 ♂ paratypes (CMNZ, NZAC, IHPC, alc), WA Makairo Stream, 27688 60814, 350 m, IMH, 8.I.2000. 2 ♂ paratypes (IHPC, alc), RI Iron Bark Hut, Maropea River, Ruahine Range 27753 61646, 550 m, IMH, 14.I.1982.

Other material examined. Larvae (IHPC, alc), RI Limestone Creek near Sixtus Lodge, 27668 61324, 560 m, IMH, 22.I.2000. Larvae (IHPC, alc), TK Te Popo Stream, 26068 62105, 820 m, IMH, 27.II.1984.

Description

Male. Wing venation cannot be distinguished from that of *A. magnicornis* (the short footstalk to apical fork 1 in Figure 5a is a variable feature). Hind wing medial veins with a distinct fork (not attenuate) at 1/4 of the wing length. Cross-vein between media and cubitus veins present. Forewing length 3.9–5.1 mm (mean = 4.4, sd = 0.30, $n = 26$).

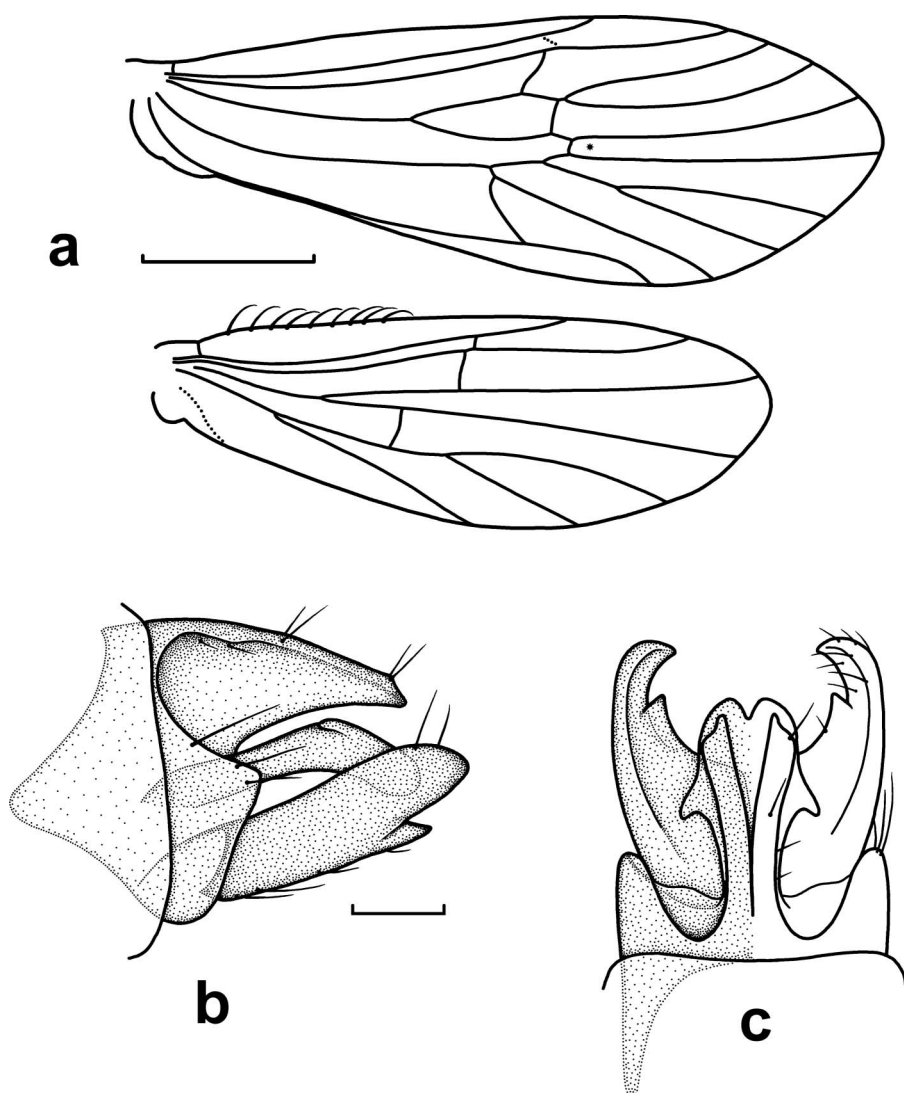


Figure 5. *Alloecentrella incisus* male (a) wings; (b, c) genitalia, lateral, dorsal.

Genitalia. Tergite X cleft almost to the base, each arm narrow and parallel-sided, medial edge projecting at the apex. Penis cover smaller than in other species of the genus, small anteriorly-directed dorso-lateral projections at mid point. Inferior appendages parallel-sided in lateral view, rounded distally and turned inwards. A small acute projection at three quarters of the length of the inferior appendage.

Female. Wing venation; hind wing fork in Rs (at base of the open radial cell) is deeper and more acute than in other species of the genus. Apical fork 2 sessile but not based with a cross-vein. Forewing length 3.5–3.6 mm ($n = 2$).

Genitalia. Ventral medial lobes are thin and divergent. A small, weakly sclerotised, blunt projection arises from an un-pigmented area medially near the apex of segment IX.

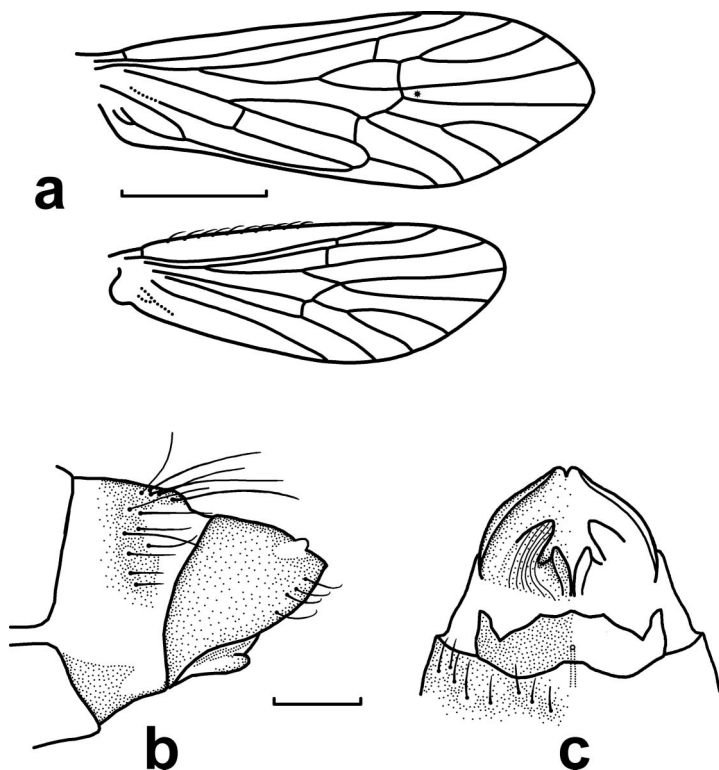


Figure 6. *Alloecentrella incisus* female (a) wings; (b, c) genitalia, lateral, dorsal.

Diagnosis

This species is similar to *A. cirratus* but the males can be distinguished by the deep split in tergite X and the lack of a scent organ on the maxillary palps. The two known female specimens are smaller than any males of the species, whereas in *A. magnicornis* and *A. linearis* females are generally 15–20% larger than males.

Distribution

Central North Island only (Wairarapa, Rangitikei, Taranaki and Hawkes Bay) (Figure 9).

Etymology

Latin *incisus*, past participle from *incidere*: to cut up, or cut into. The species is named for the deeply cleft form of tergite X.

Descriptions of the cases and larvae (Figures 7 and 8)

The larva of *A. magnicornis* was described and figured by Cowley (1978) and his description applies in general to the new species described here. The cases of all four species of *Alloecentrella* are predominantly clothed with pieces of bryophyte with only occasional mineral or other plant fragments. Leafy fragments may be overlapped like shingles or abutted flat. In some locations the plant fragments are thick and spongy but case material is not diagnostic of

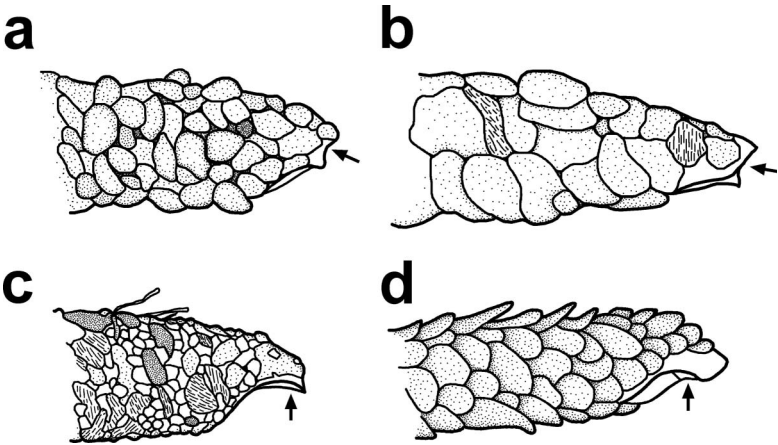


Figure 7. *Alloecentrella* larval case, posterior. (a) *A. linearis*, (b) *A. magnicornis*, (c) *A. incisus*, (d) *A. cirratus*. Aperture marked with arrow.

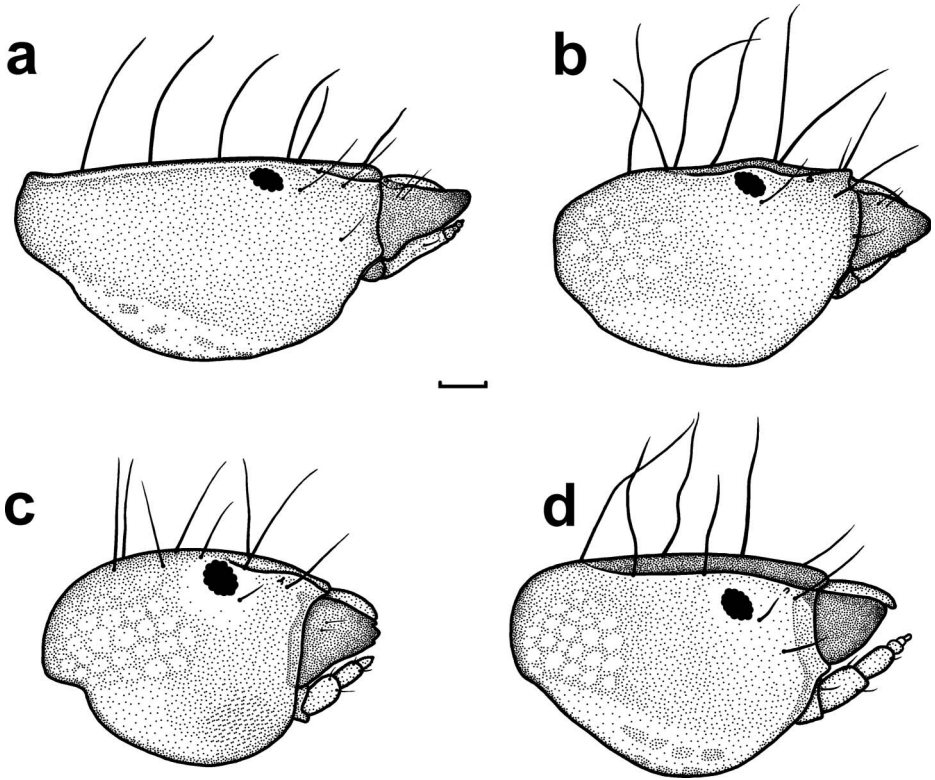


Figure 8. *Alloecentrella* larval head, lateral. (a) *A. linearis*, (b) *A. magnicornis*, (c) *A. incisus*, (d) *A. cirratus*. Scale bar = 0.1 mm.

species. The posterior aperture (Figure 7) is ventral in *A. incisus* and *A. cirratus* with a small overhanging dorsal lip (set slightly back from the posterior end in the latter). In *A. linearis* and *A. magnicornis* the aperture is terminal with a small projecting ventral lip.

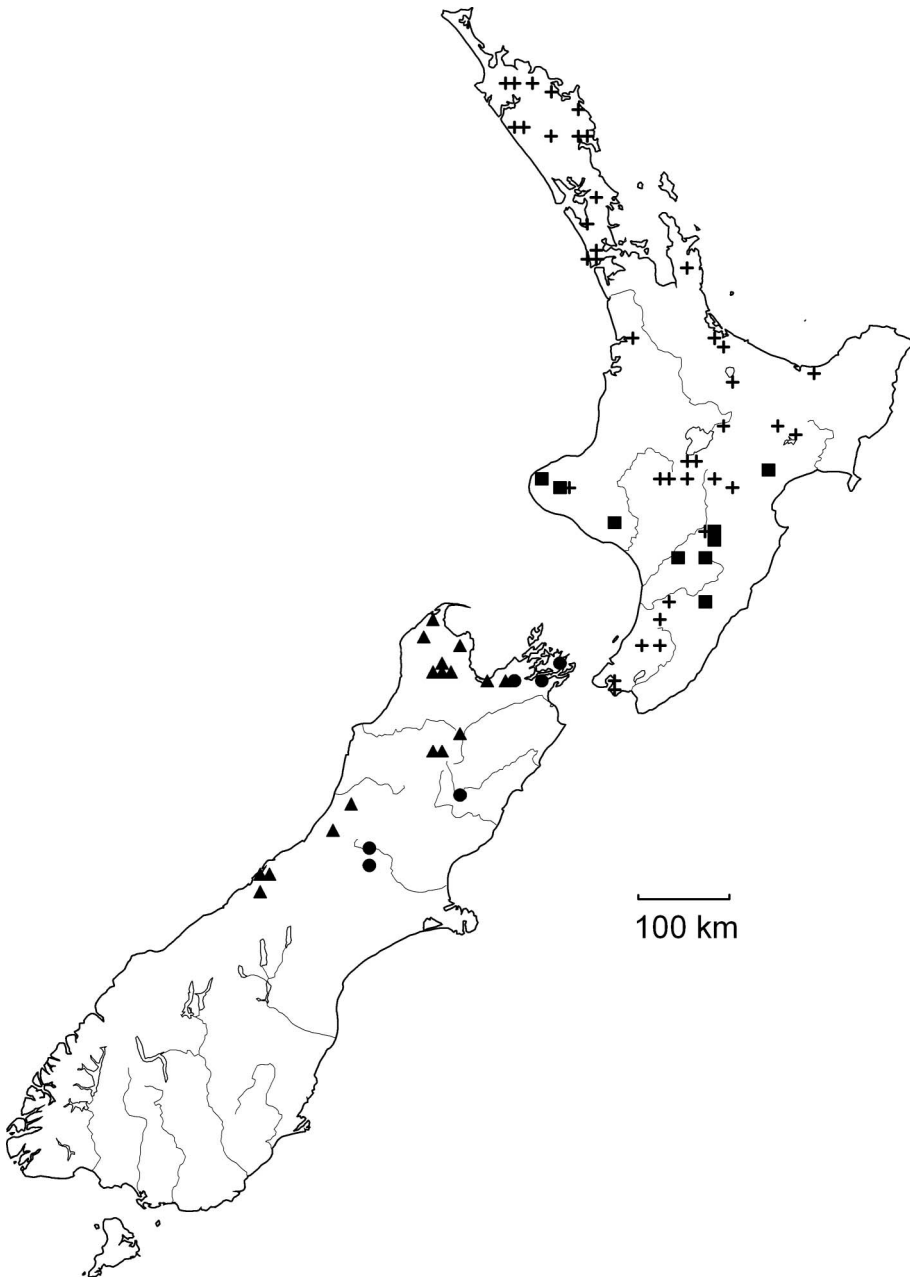


Figure 9. Distribution of *Alloeocentrella* species. *A. magnicornis*, +; *A. linearis*, ▲; *A. incisus*, ■; and *A. cirratus*, ●.

The larvae of the four species can also be distinguished by the shape of the head (Figure 8), particularly the extent of the carina. In *A. linearis* a well marked carina is continuous around the back of the head making the head appear flat with a raised rim. The carina of *A. cirratus* fades out on the vertex at about the level of the last strong black seta, the very back of the head being rounded. The carina in *A. magnicornis* is further reduced

and the back of the head is not smoothly rounded due to paired shallow depressions (Cowley 1978). *Alloecentrella incisus* has the most smoothly rounded head with carina not extending past the eye.

Phylogenetic analysis

A parsimony analysis of nine larval, two pupal and four adult characters from 20 genus or family level taxa was performed using PAUP*4.0b10 (Swofford 1999). The character matrix (Tables I and II) is adapted from Neboiss (2002) with the following changes. Character 2, “head with carina” was coded by Neboiss (2002) as with carina (1), without carina (2) and with or without carina (3). We code the third state as polymorphic. Neboiss (2002) used two binary characters for larval abdominal gills, presence and absence, and simple gills vs. branched. We list this as one 3-state character: absent, present and simple, present and branched. Character 12 coding uses the number of segments as states and is treated as polymorphic in some taxa (see Table II). Johanson and Ward (2001, 2002) and de Moor (1993) consider the large, slightly curved macrotrichia on the hind wing costal margin of Helicophidae as hamuli but Neboiss (2002) considers only Helicopsychidae to have true (strongly hooked) hamuli. In character 15 we have not distinguished hamuli and macrotrichia as different wing coupling mechanisms. Multistate characters were treated as ordered except for characters numbered 1 and 10. We have also made minor corrections based on information in Scott (1993), added five taxa to the matrix based on our observations of specimens of the genera *Alloecentrella*, *Zelolessica*, *Pycnocentrella* and family Chathamiidae, and separately coded the two Australian genera of Helicophidae based on additional information in Neboiss (2002). Trees are displayed using Sericostomatidae as the outgroup. In a phylogenetic analysis encompassing most families of Trichoptera using morphological and molecular characters (Kjer et al. 2001), Sericostomatidae and Beraeidae were basal to the other families we consider here.

Table I. Coded character states for Helicophidae and related taxa. Based on Neboiss (2002).

Larvae	
1. Case material	sand only (1), silk only (2), mixed sand, silk and/or plant (3)
2. Head carina	present (1), absent (2)
3. Antennae	near base of mandible (1), halfway between eye and base of mandible (2), close to eye (3)
4. Ventral apotome shape	equilateral triangle (1), elongate triangle (2), quadrangular (3)
5. Genae separated	only anteriorly (1), entirely separated or abutting (2)
6. Abdominal dorsal hump	present (1), reduced or absent (2)
7. Lateral humps	with spiny patch (1), without spiny patch or sclerites (2)
8. Abdominal gills	absent (1), present and simple (2), present and branched (3)
9. Anal claw accessory hooks	one (1), two or three (2), dorsal comb (3)
Pupae	
10. Mandible inner margin	finely serrate (1), dentate (2), broad basally, tapering to slender apical process or seta (3)
11. Abdominal dorsal plates	on segments 2–6 (1), on segments 3–6 (2), without plates (3)
Adults	
12. Male maxillary palps segments	one (1), two (2), three (3), four (4), five (5)
13. Pronotum warts	two pairs (1), one pair (2), single fused (3)
14. Forewing radial cell	closed (1), open (2), absent (3)
15. Wing coupling	none, or frenular hairs at base (1), macrotrichia or hamuli (2)

Table II. Data matrix for character states of Helicophidae and related taxa. Characters polymorphic within a taxon are shown in {} or, if all states present, as “?”.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Sericostomatidae	1	?	2	2	1	2	1	2	1	1	2	{1,2,3,5}	2	1	2
Hydrosalpingidae	2	2	1	1	1	1	1	1	1	2	1	5	2	1	2
Petrothrincidae	1	2	1	3	2	1	2	3	3	1	2	5	1	1	2
Barbarochthonidae	3	1	1	3	2	1	1	1	2	1	1	3	3	1	2
Helicopsychidae	1	1	2	1	1	2	2	{1,3}	3	1	2	{2,3,4}	2	1	2
Beraeidae	1	1	2	3	2	1	2	3	1	1	3	5	2	3	2
Antipodoeciidae	1	1	2	1	2	2	1	1	1	3	2	3	2	2	1
Anomalopsychidae	1	1	2	1	2	2	2	1	3	3	2	{4,5}	1	1	2
Conoesucidae	?	?	2	3	2	2	1	3	1	1	2	{1,2,3}	2	{1,3}	2
Chathamiidae	3	2	2	3	2	1	1	3	1	1	2	5	2	1	2
Helicophidae:															
<i>Alloecella</i>	3	?	2	2	2	1	1	1	1	2	2	5	2	2	2
<i>Helicopha</i>	3	?	2	2	2	1	1	1	1	2	2	5	2	1	2
<i>Alloecentrella</i>	3	?	2	2	2	1	1	1	1	1	1	5	3	1	2
<i>Zelolessica</i>	3	?	2	2	2	1	1	1	1	1	1	5	2	3	2
<i>Heloccabus</i>	2	1	1	2	2	2	1	2	2	1	1	3	2	1	1
Calocidae:															
<i>Caloca</i>	1	2	3	1	1	1	1	1	1	1	2	5	1	1	2
<i>Caenota</i>	3	2	3	1	1	1	1	2	1	1	2	5	{1,2}	1	2
<i>Phiocaloca</i>	1	2	2	1	1	1	1	1	1	1	2	5	1	1	2
<i>Tamasia</i>	1	1	3	1	1	1	1	1	1	1	2	4	1	1	2
<i>Pycnocentrella</i>	1	1	2	1	1	1	2	1	1	1	2	5	1	1	1

Results

Parsimony analysis of the data in Table II by branch-and-bound search and assuming ordered multistate characters, resulted in six trees of length 64. Figure 10 shows the semi-strict consensus tree. The sister-group of *Alloecentrella* is the New Zealand genus *Zelolessica* and together they are grouped with the two Australian genera in the Helicophidae (*Alloecella* and *Helicopha*). These four genera were monophyletic in three of the trees but grouped in a trichotomy with Chathamiidae, Beraeidae and Petrothrincidae in the other three trees. Treating all multi-state characters as unordered resulted in 589 most parsimonious trees from a heuristic search. Although there is much less resolution with this second analysis, the arrangement of *Alloecentrella* within the Helicophidae is still supported. *Alloecentrella* grouped with *Zelolessica*, *Alloecella*, *Helicopha*, Barbarochthonidae and Chathamiidae in the majority rule consensus tree, not with any members of the Calocidae.

Discussion

Parsimony analysis clearly indicates that *Alloecentrella* should be placed in the family Helicophidae rather than Calocidae. There are two diagnostic characters in this respect: in the larva the ventral apotome is an equilateral triangle in Calocidae but more elongate, resulting in complete separation of the genae in Helicophidae. In the adult, there are two pairs of setal warts on the pronotum in Calocidae but either a single pair or a fused single wart in Helicophidae. Within the genus *Alloecentrella* a range of degrees of fusion can be seen with two abutting warts in *A. incisus* and *A. cirratus* but a single wart with only a faint constriction marking the fusion in *A. magnicornis* and *A. linearis*. A further supporting

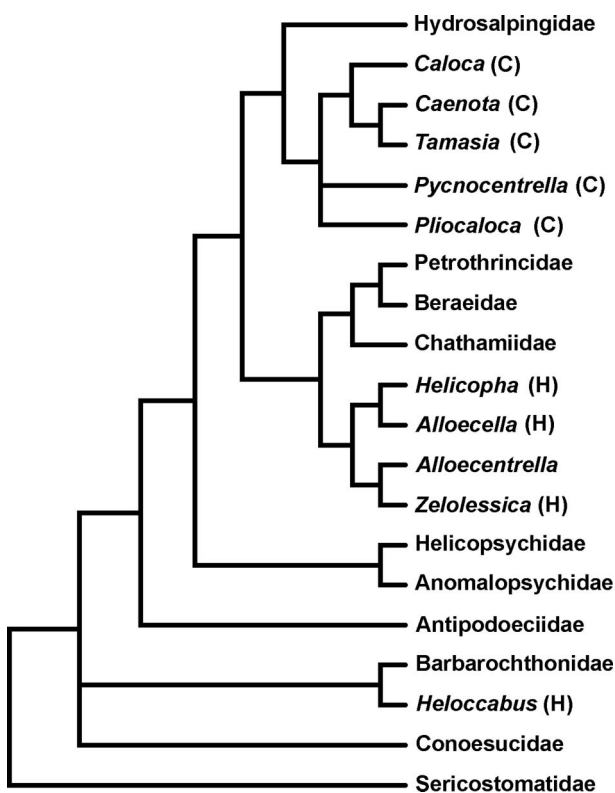


Figure 10. Phylogeny of Helicophidae and related taxa, consensus of the six equally parsimonious trees for the data in Table II. Present family placement of genera indicated by (C): Calocidae and (H): Helicophidae.

character is the shape of the scutellum warts, an elongated pair in the Calocidae but a single or pair of round warts in Helicophidae (Neboiss 1986). In *Alloecentrella* there is a single round scutellum wart.

The results of phylogenetic analyses of families in the Sericostomatoidea by de Moor (1993), Kjer et al. (2001) and Neboiss (2002) have little in common and our analysis does not support any of these previous arrangements. However, our intention was simply to establish the correct family placement of *Alloecentrella*, not to provide a phylogenetic classification of the Sericostomatoid families which will clearly require much wider sampling of characters and taxa to obtain a robust result.

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